TECHNICAL DATASHEET · INDEPENDENT DOCUMENT

Airton serial protocol

Serial link between the air-conditioner board and its Wi-Fi module — standalone reference for a local replacement.

WARNING

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1. Overview

The bus is a clear-text serial link with two data wires and a common ground. The Wi-Fi module is the master (`0x21`); the air-conditioner board (`0xD5`) and the display (`0xD1`) are slaves. The module polls the board, forwards changes, and maintains a separate link with the display.

Line	Direction and frames
A	module `0x21` → board `0xD5` / display `0xD1`: `A1`, `A2`, `AB`, `A5`
B	board `0xD5` / display `0xD1` → module `0x21`: `A3`, `A6`, `AC`, `A4`

Physical layer	Value
Bitrate	9600 baud, 8N1
Levels	TTL 5 V
Wires	TX, RX, common GND
Master	`0x21` — module

Address	Role
`0x21`	Wi-Fi module (master)
`0xD5`	air-conditioner board
`0xD1`	display / thermostat

2. Frame and integrity

```
7A 7A | SRC | DST | LEN | 00 00 | TYPE | PAYLOAD | CRC16
```

`SRC`/`DST`: source and destination addresses. `LEN`: total length including CRC (12–34). `TYPE`: message type. `CRC16`: CRC-16/MODBUS, high byte first, computed over `0 .. LEN-3`.

CRC-16/MODBUS	Value
Polynomial	`0x8005` reflected → `0xA001`
Init	`0xFFFF`
RefIn / RefOut	yes / yes
Final XOR	`0x0000`
Coverage	`0 .. LEN-3`
Byte order	high byte first

Test vector

```
7A 7A 21 D5 0C 00 00 AB 0A 0A  →  FC F9
```

Frames are split by `LEN`: find `7A 7A`, read the length, wait for the complete frame, verify the CRC, then consume.

3. Messages

Type	Size	Direction	Cadence	Role
`AB`	12	`21 → D5`	60 s	module poll to the board
`A2`	12	`21 → D5`	at boot	startup poll
`AC`	18	`D5 → 21`	80 ms	poll response: total usage time and revision
`A1`	24	`21 → D5`	on change	module command to the board
`A3`	34	`D5 → 21`	60 s + immediate	full board state
`A6`	28	`D5 → 21`	with `A3`	complementary state, all zero
`A5`	27	`21 → D1`	10 s	module announcement to the display
`A4`	13	`D1 → 21`	0.1 s	display acknowledgement

4. Implementation

- Verify the voltage (TTL 5 V) and the common ground before connecting.
- Power the ESP from the module connector's +12 V through a 5 V regulator (a 12 V → 5 V buck module or a bare 7805); never connect +12 V directly to the ESP.
- Never transmit on the bus while the original module is still connected.
- Preserve unknown fields and verify the CRC of every frame.
- Wait for the first `A3` before sending `A1`; reject frames under 10 bytes and release a partial frame after 50 ms.
- Test in listen-only mode first, then transmit, with a rollback path.

5. A1 command — module → board

Offset	Field	Value
8–11	reserved	00 00 00 00
12	state	bit 7 super; bits 4–6 fan; bit 3 power; bits 0–2 mode
13	setpoint	bits 0–4 = setpoint − 16; bit 5 = °F; bit 6 = quiet
14	swing	bit 0 vertical, bit 4 horizontal; preserve other bits
15	flags	see the flags table
16–21	reserved	00
22–23	CRC16	high byte first

Example — heating, 24 °C, auto, vane off:

```
7A 7A 21 D5 18 00 00 A1 00 00 00 00 0C 08 00 80 00 00 00 00 00 00 8A 98
```

6. A3 state — board → module

Offset	Field	Value
8–9	prefix	0A 0A
10	room temperature	integer °C
11	room temperature decimal	tenths of °C
13	state	same as A1[12]
14	setpoint	same as A1[13]
15	swing	bit 0 vertical, bit 4 horizontal
16	flags	see the flags table
21	category	01 cool/heat_cool, 02 dry, 03 fan_only, 04 heat
23–24	revision	counter, e.g. BC 58
25–31	identifier	DB11370 (ASCII)
32–33	CRC16	high byte first

7. Encodings

Mode bits	Mode
0	heat_cool
1	cool
2	dry
3	fan_only
4	heat

Fan bits	Fan speed
0	auto (or quiet via setpoint bit 0x40)
1	low
2	medlow
3	medium
4	medhigh
5	high
6	no fan mode; preserved
8	super bit, combined with a speed
D	strong / TURBO (8 + 5)

Mode 0–4 are in the low 3 bits with the power bit 0x08; off clears the power bit and keeps the mode bits. **Swing**: bit 0 vertical, bit 4 horizontal; 00 off, 01 vertical, 10 horizontal, 11 both; other bits preserved. Positions 1–5 are not exposed.
Flags (A1[15]/A3[16]): 0x80 display on, 0x40 ionizer, 0x10 health (anti-mould/aux heat), 0x08/0x04 unknown/status, 0x02 sleep, 0x01 eco. Flags are preserved in A1; cooling uses 0x80.

8. AC poll response

AC[8–9] = 0A 0A; AC[10–11] = total usage time in hours (big-endian, DP 102); AC[13–14] = revision (matches A3[23:24]); AC[12] and AC[15] are unknown; AC[16–17] = CRC16. 0x5AD1 = 23249 h, BC75 is the revision.

9. Timing and sequences

Event	Frames	Delay
display heartbeat	A5 then A4	10 s
board poll	AB then AC	60 s
periodic state	A3 + A6	60 s
command	A1 then A3 + A6	170 ms
remote key press	A3 + A6	immediate, no A1

At power-up the module sends an A5 variant (byte 11 04, bytes 16–18 01 01 01), then an A2 startup poll, then the normal A5. After the power cycle A5[16] becomes 01; its meaning is not confirmed.

10. Reference frames

```
AB  7A 7A 21 D5 0C 00 00 AB 0A 0A FC F9
A2  7A 7A 21 D5 0C 00 00 A2 0A 0A FE 29
AC  7A 7A D5 21 12 00 00 AC 0A 0A 5A D1 00 BC 75 01 78 27
A1  7A 7A 21 D5 18 00 00 A1 00 00 00 00 0C 08 00 80 00 00 00 00 00 00 8A 98
A3  7A 7A D5 21 22 00 00 A3 0A 0A 19 00 00 0C 08 01 98 00 00 00 00 04 64 BC 58 44 42 31 31 33 37 30 76 9E
```

Limitations — Vane positions 1–5, some display/status bits and fields not listed here are not documented; preserve unknown values. Trademarks belong to their respective owners. This document is neither official nor approved by Airton or Tuya.

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